

Ayush Kumar

📍 Meerut India ✉ ayush2003kumar19@gmail.com ☎ 6397752343 in Ayush Kumar 🌐 Ayush Kumar

Education

- National Institute of Technology, Srinagar** *Mar 2021 – Jul 2025*
BTech in Computer Science, GPA: 7.46
- Jawahar Navodaya Vidyalaya, Kathua** *Jun 2018 – Jun 2020*
Higher Secondary Education (Non-Medical), Percentage: 90%
- Jawahar Navodaya Vidyalaya, Kathua** *Apr 2016 – Apr 2018*
Senior Secondary Education, Percentage: 78.4%

Experience

- AI & ML Engineer — Rashmi Metaliks Limited** *Jan 2025 – Present*
- Engineered an AI-driven CRM platform using **Gemini AI, React.js, Prisma, and PostgreSQL**, automating **60+ business workflows** through **AI agents, multi-step reasoning pipelines, forecasting modules, and an MCP-based chatbot** with tool-calling capabilities.
 - Led the development of a **scalable full-stack system** supporting **1,300+ leads valued at \$1B+** across **1,000+ users**, implementing **real-time activity tracking with Socket.IO**, MIS reporting, and interactive business intelligence dashboards.
 - Developed an **NLP-based resume parsing engine** leveraging information extraction techniques and **500+ domain-specific heuristics** to convert unstructured resumes into structured candidate profiles, reducing **manual screening effort by 70%**.
 - Built a **real-time candidate data ingestion pipeline** using **IMAP integration** to process **1,000+ applications**, enabling automated document intelligence, data normalization, and structured storage in **PostgreSQL** for downstream analytics and recruitment workflows.

Projects

- Real-Time Car Number Plate Extraction (ANPR)** *Apr 2026 – May 2026*
- Designed and deployed a **real-time ANPR system** using **Computer Vision and Deep Learning** to detect and extract vehicle number plates from **live video streams**.
 - Applied **OpenCV-based preprocessing** and **YOLO object detection** techniques to achieve **high-precision number plate localization**.
 - Integrated **Tesseract OCR** for accurate **alphanumeric character recognition**, enabling applications such as **toll automation** and **intelligent traffic monitoring**.
- House Price Prediction using Machine Learning** *Jan 2026 – Mar 2026*
- Developed a **supervised machine learning regression model** to predict **house prices** using structured **real-estate data**, incorporating **data preprocessing, EDA, and feature engineering**.
 - Evaluated and compared multiple models, including **Linear Regression** and **Random Forest**, achieving an **R² score above 0.90**.
 - Improved model performance through **hyperparameter tuning** and **cross-validation**, leveraging **RMSE** and **R²** metrics for optimization and validation.

Technologies & Skills

- Languages:** Python, SQL.
- AI & ML:** ML, DL, NLP, LLMs, RAG, AI Agents, Model Evaluation.
- Frameworks:** FastAPI, LangChain, LangGraph.
- Tools & Database:** Git, Docker, ChromaDB, PostgreSQL.
- Core CS Concepts:** DSA, DBMS, OOP, Operating Systems, Computer Networks.